The q-characters and Jacobi-Trudi identity

1. Let $g = sl_{n+1}$ and $U_q(\hat{g}) = \langle x_{i,r}^{\pm}, k_i^{\pm 1}, \phi_{i,\pm r}^{\pm} \rangle$ be at level 0. Set $I = \{1, \dots, n\}$.

For $a \in \mathbb{C}^\times$, $i \in I$, let $Y_{i,a}$ be the I-tuple of rational functions:

$$Y_{i,a} := \left(1, \dots, 1, q\,\frac{1 - aq^{-1}z}{1 - aqz}, 1, \dots, 1 \right) \quad (i\text{-th fundamental weight})$$

Let $\mathcal{P}$ be the free group generated by the $Y_{i,a}$'s with product given coordinate-wise and let $\mathcal{P}^+$ be the set of dominant monomials.

For $m \in \mathcal{P}$, say: $m = (m_1, m_2, \dots, m_n)$, we write

$$m_i^{\pm}(z) = \sum_{r=0}^{\infty} m_{i,\pm r}^{\pm} z^{\pm r} \in \mathbb{C}[[z^{\pm 1}]]$$

to denote the power series expansion of the i-th rational function of m at $z = 0$ and $z = +\infty$, respectively.

Let V be a finite-dimensional $U_q(\hat{g})$-module. The l-weight space of $m \in \mathcal{P}$ is the common generalized eigenspace:

$$V_m = \left\{ v \in V : \forall i \in I, n \geq 0, \left(\phi_{i,\pm r}^{\pm} - m_{i,\pm r}^{\pm} \right)^N = 0, N >> 0 \right\}.$$

Let $\mathbb{Z}\mathcal{P} := \mathbb{Z}[Y_{i,a}^{\pm 1}]_{i \in I, a \in \mathbb{C}^\times}$. The q-character of V is the element of $\mathbb{Z}\mathcal{P}$:

$$\chi_q(V) := \sum_{m \in \mathcal{P}} (\dim V_m)\, m.$$

The isomorphism classes of irreducible finite-dim'l modules are in bijection with $\mathcal{P}^+$:

$$\hat{m} \in \mathcal{P}^+ \leftrightarrow [V(\hat{m})],$$

where $V(\hat{m})$ is the highest ℓ-weight module cyclic on v_0 satisfying:

$$\phi^{\pm}_{i,\pm n} \cdot v_0 = \hat{m}^{\pm}_{i,\pm n} v_0, \quad x^+_{i,r} \cdot v_0 = 0.$$

Let $X_{i,a} := \chi_q(V(Y_{i,a}))$. Our objective is to give an explicit description of the q-characters of evaluation modules as polynomials in

$$\mathbb{Z}[X_{1,a}, X_{2,a}, \dots, X_{n,a}]_{a \in \mathbb{C}^\times}.$$

2. Let $S = \{1, \dots, n+1\}$. Let $\mathbb{Z}\Lambda$ be the commutative ring over $\mathbb{Z}$ generated by the variables $\{\Lambda_{i,a}\}_{i \in S, a \in \mathbb{C}^\times}$ satisfying the relation:

$$\Lambda_{1,aq^{-2}} \Lambda_{2,aq^{-4}} \Lambda_{3,aq^{-6}} \cdots \Lambda_{n+1,aq^{-2(n+1)}} = 1.$$

Prop: $\mathbb{Z}\mathcal{P} = \mathbb{Z}[Y^{\pm 1}_{i,a}]_{i \in I, a \in \mathbb{C}^\times} \simeq \mathbb{Z}\Lambda = \mathbb{Z}[\Lambda_{i,a}]_{i \in S, a \in \mathbb{C}^\times}$ as rings.

Pf. Set $Y_{0,a} := 1$, $Y_{n+1,a} := 1$ $\forall a \in \mathbb{C}^\times$. Define $\varphi : \mathbb{Z}\Lambda \to \mathbb{Z}\mathcal{P}$ by:

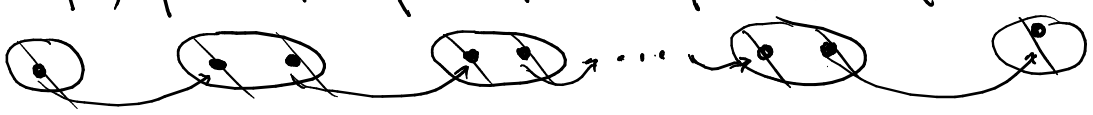
$$\Lambda_{i,a} \mapsto Y^{-1}_{i-1,aq^i} Y_{i,aq^{i-1}} \quad (\Lambda_{1,a} \mapsto Y_{1,a}, \ \Lambda_{n+1,a} \mapsto Y^{-1}_{n,aq^{n+1}}).$$

We verify φ is well-defined: $\prod_{i=1}^{n+1} \varphi(\Lambda_{i,aq^{-2i}}) = 1$.

Note: if $b = aq^{-2}$, then $\varphi(\Lambda_{i,a})\varphi(\Lambda_{i+1,b}) = Y^{-1}_{i-1,aq^{i+1}} Y_{i,aq^{i-1}} Y^{-1}_{i,bq^{i+1}} Y_{i+1,bq^i}$

That is, the positive degree term of $\varphi(\Lambda_{i,a})$ cancels out the negative degree term of $\varphi(\Lambda_{i+1,b})$. Hence,

$$\varphi(\Lambda_{1,aq^{-2}})\varphi(\Lambda_{2,aq^{-4}})\varphi(\Lambda_{3,aq^{-6}}) \cdots \varphi(\Lambda_{n,aq^{-2n}})\varphi(\Lambda_{n+1,aq^{-2(n+1)}}) = 1.$$



Now, we define $\psi : \mathbb{Z}\mathcal{P} \to \mathbb{Z}\Lambda$ by:

$$Y_{i,a} \mapsto \prod_{k=1}^{i} \Lambda_{k,a q^{i-2k+1}}$$

$$Y_{i,a}^{-1} \mapsto \prod_{k=i+1}^{n+1} \Lambda_{k,a q^{i-2k+1}}.$$

It's immediate that ψ is well-defined. We show $\varphi \circ \psi = id$, $\psi \circ \varphi = id$.

- $\varphi \circ \psi = id$:

$\varphi(\psi(Y_{i,a})) = \prod_{k=1}^{i} \varphi(\Lambda_{k,a q^{i-2k+1}})$. Using the same reasoning as before in computing successive products with shift q^{-2}, we find that the RHS equals the positive degree term of $\varphi(\Lambda_{i,a q^{-i+1}})$, which is $Y_{i,a q^{i-1-i+1}} = Y_{i,a}$.

Analogous argument shows that $\varphi(\psi(Y_{i,a}^{-1})) = Y_{i,a}^{-1}$ as well.

- $\psi \circ \varphi(\Lambda_{i,a}) = \psi(Y_{i-1,a q^{i}}^{-1}) \psi(Y_{i,a q^{i-1}})$

$$= \prod_{k=i}^{n+1} \Lambda_{k,a q^{i+i-1-2k+1}} \prod_{k=1}^{i} \Lambda_{k,a q^{i-1+i-2k+1}}$$

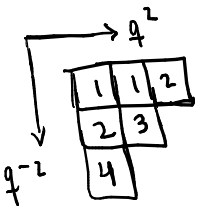
$$= \Lambda_{i,a} \prod_{k=1}^{n+1} \Lambda_{k,q^{2(i-k)}} = \Lambda_{i,a}.$$

∎

3. For a partition λ, let $SSYT(\lambda, S)$ be the set of semi-standard Young Tableau of shape λ and fillings in the set $S = \{1, 2, \dots, n+1\}$. We denote by $f_{ij} \in S$ the filling of the box at the i-th row and j-th column of T.

For a Tableau $T \in SSYT(\lambda, S)$ and $a \in \mathbb{C}^\times$, define Λ_a^T by:

$$\Lambda_a^T := \prod_{i,j} \Lambda_{f_{ij}, aq^{2(j-i)}}.$$

<u>Ex.</u> $n+1=4$, $T = $ 

, $\Lambda_a^T = \Lambda_{1,a}\Lambda_{2,aq^{-2}}\Lambda_{4,aq^{-4}}\Lambda_{1,aq^2}\Lambda_{3,a}\Lambda_{2,aq^2}$. ◈

↳ so that we can define a $\mathfrak{sl}_{n+1}$-weight.

For λ partition with $\ell(\lambda) \leq n$, let $\hat{T}$ be the highest weight tableau given by the filling $f_{ij} = i \ \forall \ i,j$.

<u>Lemm.</u> We have $\varphi(\Lambda_a^{\hat{T}}) = \prod_{j=1}^{\lambda_1} Y_{\lambda_j', a(j)} \in \mathcal{P}_+$ where $a(j) = aq^{2j-\lambda_j'-1}$.

Pf. The j-th column of $\hat{T}$ is exactly: 

Thus, $\Lambda_a^{\hat{T}} = \prod_{j=1}^{\lambda_1} \Lambda_{1,aq^{2(j-1)}}\Lambda_{2,aq^{2(j-1)-2}}\Lambda_{3,aq^{2(j-1)-4}}\cdots\Lambda_{\lambda_j',aq^{2(j-1)-2(\lambda_j'-1)}}$.

$$\Rightarrow \quad \varphi(\Lambda_a^{\hat{T}}) = \prod_{j=1}^{\lambda_1} \varphi(\Lambda_{1,aq^{2(j-1)}})\cdots\varphi(\Lambda_{\lambda_j',aq^{2(j-1)-2(\lambda_j'-1)}})$$

cancelling out successive terms ↳ $= \prod_{j=1}^{\lambda_1} Y_{\lambda_j', aq^{2j-2-2\lambda_j'+2+\lambda_j'-1}} = \prod_{j=1}^{\lambda_1} Y_{\lambda_j', a(j)}$. ∎

For λ with $\ell(\lambda) \leq n$, we denote by $V(\lambda, a) := V(\varphi(\Lambda_a^{\hat{T}}))$ the irreducible finite dim'l module of highest ℓ-weight $\varphi(\Lambda_a^{\hat{T}})$.

Thm. Let λ with $\ell(\lambda) \le n$. Then, $V_a(\lambda)$ is isomorphic to the evaluation module of $V(\lambda)$ at a. Moreover, we have:

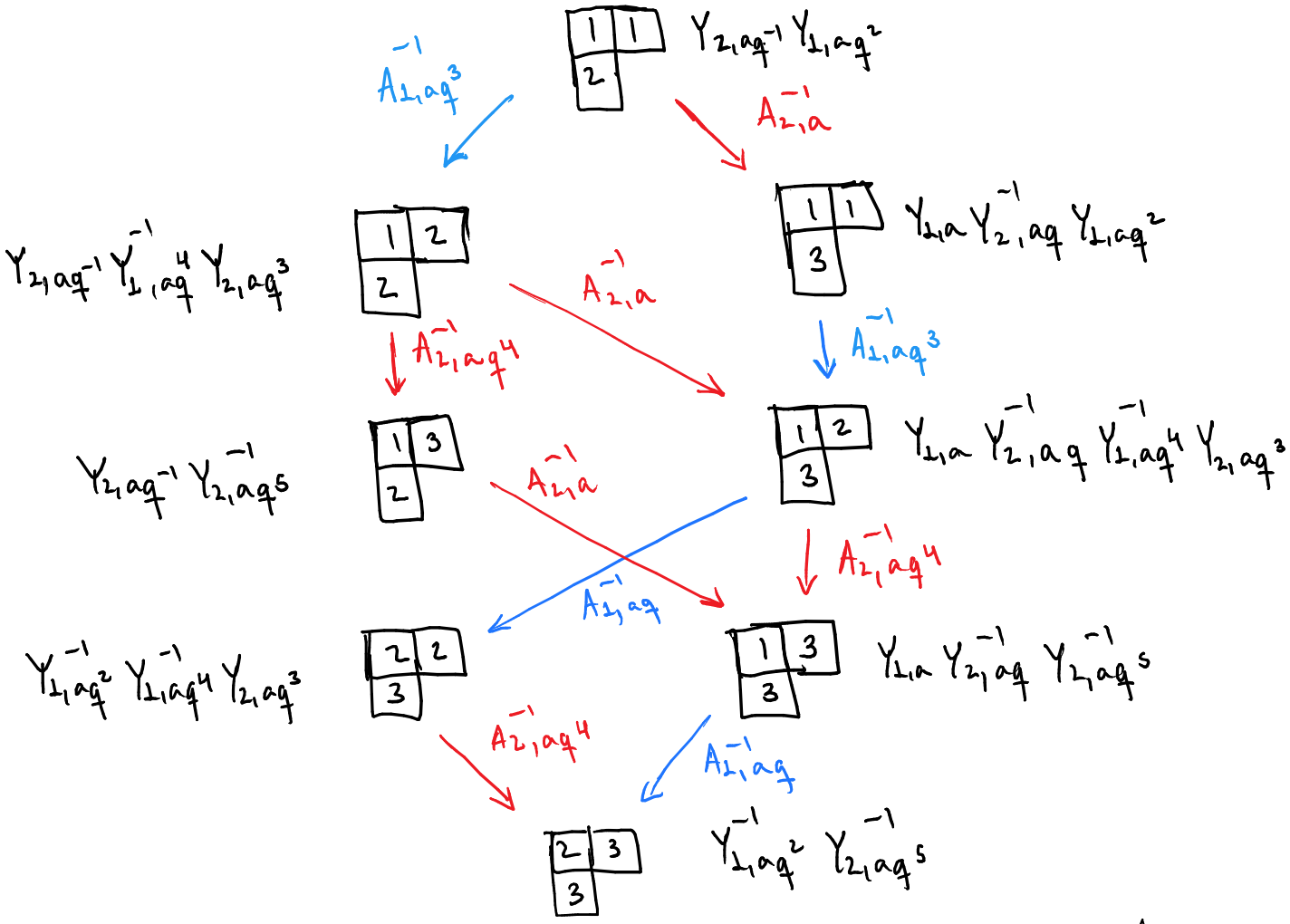
$$\chi_q(V(\lambda,a)) = \sum_{T \in SSYT(\lambda,S)} \psi(\Lambda_a^T).$$

Proof is set to be on hold for now. However, we illustrate this theorem with an explicit example.

Ex. $n+1=3$, $\lambda=(2,1)$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \rightsquigarrow \begin{aligned} A_{1,b}^{-1} &= Y_{1,bq^{-1}}^{-1} Y_{1,bq} Y_{2,b} \\ A_{2,b}^{-1} &= Y_{2,bq}^{-1} Y_{2,bq} Y_{1,b} \end{aligned}$$

$$\boxed{1} = Y_{1,a}$$
$$\boxed{2} = Y_{1,aq^2}^{-1} Y_{2,aq}$$
$$\boxed{3} = Y_{2,aq^3}^{-1}$$



4. Let R be the non-commutative ring generated by $\mathbb{Z}\Lambda$ and $\mathbb{Z}[[X]]$ with relations: $X\Lambda_{i,a} = \Lambda_{i,aq^{-2}} X$ for $i \in S$, $a \in \mathbb{C}^\times$.

Define $E_a(\Lambda, X), H_a(\Lambda, X) \in R$ as follows:

- $E_a(\Lambda, X) = \overrightarrow{\prod_{1 \le k \le n+1}} (1 + \Lambda_{k,a} X) = (1 + \Lambda_{1,a} X) \cdots (1 + \Lambda_{n+1,a} X)$
- $H_a(\Lambda, X) = \overleftarrow{\prod_{1 \le k \le n+1}} (1 - \Lambda_{k,a} X)^{-1} = (1 - \Lambda_{n+1,a} X)^{-1} \cdots (1 - \Lambda_{1,a} X)^{-1}$.

Let $e_{i,a}, h_{i,a} \in \mathbb{Z}\Lambda$ be so that

$$E_a(\Lambda, X) = \sum_{i=0}^{\infty} e_{i,a} X^i \quad \text{and} \quad H_a(\Lambda, X) = \sum_{i=0}^{\infty} h_{i,a} X^i.$$

Note that $e_{i,a} = 0$ for $i > n+1$.

Prop. We have: $e_{k,a} = \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n+1} \Lambda_{i_1,a} \Lambda_{i_2,aq^{-2}} \Lambda_{i_3,aq^{-4}} \cdots \Lambda_{i_k,aq^{-2(k-1)}}$,

$$h_{k,a} = \sum_{1 \le i_1 \le i_2 \le \cdots \le i_k \le n+1} \Lambda_{i_1,aq^{-2(k-1)}} \Lambda_{i_2,aq^{-2(k-1)+2}} \Lambda_{i_3,aq^{-2(k-1)+4}} \cdots \Lambda_{i_k,a}.$$

Pf.

We start with $e_{k,a}$. As we multiply out $E_a(\Lambda, X) = (1 + \Lambda_{1,a} X) \cdots (1 + \Lambda_{n+1,a} X)$, only the products of the form $\Lambda_{i_1,a} X \Lambda_{i_2,a} X \cdots \Lambda_{i_k,a} X$ can contribute to the coefficient of X^k after we apply the relations of R. Indeed, we get that $\Lambda_{i_1,a} X \cdots \Lambda_{i_k,a} X = \Lambda_{i_1,a} \Lambda_{i_2,aq^{-2}} \Lambda_{i_3,aq^{-4}} \cdots \Lambda_{i_k,aq^{-2(k-1)}} X^k$. By summing over all such products, the expression for $e_{k,a}$ follows.

Now, we work with $h_{k,a}$. First, notice that $(1-\Lambda_{i,a}X)^{-1} = \sum_{j=0}^{\infty}(\Lambda_{i,a}X)^j$. By induction, we easily see that $(\Lambda_{i,a}X)^j = \Lambda_{i,a}\Lambda_{i,a}q^{-2}\cdots\Lambda_{i,a}q^{-2(j-1)}X^j$. Thus,

$$(1-\Lambda_{i,a}X)^{-1} = \sum_{j=0}^{\infty}\Lambda_{i,a}\Lambda_{i,a}q^{-2}\cdots\Lambda_{i,a}q^{-2(j-1)}X^j.$$

Now, let us multiply out $H_a(\Lambda,X) = (1-\Lambda_{n+1,a}X)^{-1}\cdots(1-\Lambda_{1,a}X)^{-1}$. The coefficient of X^k is coming from products of the form:

$$\left(\Lambda_{i_1,a}\cdots\Lambda_{i_1,a}q^{-2(j_1-1)}X^{j_1}\right)\left(\Lambda_{i_2,a}\cdots\Lambda_{i_2,a}q^{-2(j_2-1)}X^{j_2}\right)\cdots\left(\Lambda_{i_s,a}\cdots\Lambda_{i_s,a}q^{-2(j_s-1)}X^{j_s}\right),$$

where $1\le i_s<\cdots<i_1\le n+1$ and $j_1+\cdots+j_s=k$. By using the relations of $\mathcal{R}$, we find that the above is equal to:

$$\Lambda_{i_1,a}\cdots\Lambda_{i_1,a}q^{-2(j_1-1)}\Lambda_{i_2,a}q^{-2j_1}\cdots\Lambda_{i_2,a}q^{-2(j_1+j_2-1)}\cdots\Lambda_{i_s,a}q^{-2(j_1+\cdots+j_{s-1})}\cdots\Lambda_{i_s,a}q^{-2(k-1)}X^k.$$

By setting $i_1'=\cdots=i_{j_s}'=i_s$, $\ldots$, $i_{j_{s+\cdots+j_2+1}}'=\cdots=i_k'=i_1$, we rewrite it as

$$\Lambda_{i_1',a}q^{-2(k-1)}\Lambda_{i_2',a}q^{-2(k-1)+2}\Lambda_{i_3',a}q^{-2(k-1)+4}\cdots\Lambda_{i_k',a}X^k,$$

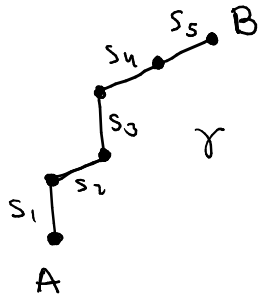
with $1\le i_1'\le i_2'\le\cdots\le i_k'\le n+1$. By summing over all such products, the coefficient of X^k is exactly $h_{k,a}$. ∎

<u>Coroll.</u> We have $e_{k,a} = \sum_{T\in SSYT(1^k,S)}\Lambda_a^T$ and $h_{k,a} = \sum_{T\in SSYT((k),S)}\Lambda_a^T q^{-2(k-1)}$. In partialar, $\varphi(e_{k,a}) = \chi_q(V(1^k,a))$ $(k\le n)$ and $\varphi(h_{k,a}) = \chi_q(V(k,aq^{-2(k-1)}))$.

Pf. $T\in SSYT(1^k,S)$ is of the form $\downarrow q^{-2}$ $\begin{array}{|c|}\hline i_1\\\hline i_2\\\hline \vdots\\\hline i_k\\\hline\end{array}$ with $1\le i_1<\cdots<i_k\le n+1$.

Then, $\Lambda_a^T = \Lambda_{i_1,a}\Lambda_{i_2,a}q^{-2}\cdots\Lambda_{i_k,a}q^{-2(k-1)}$. The proof for $h_{k,a}$ is similar. ∎

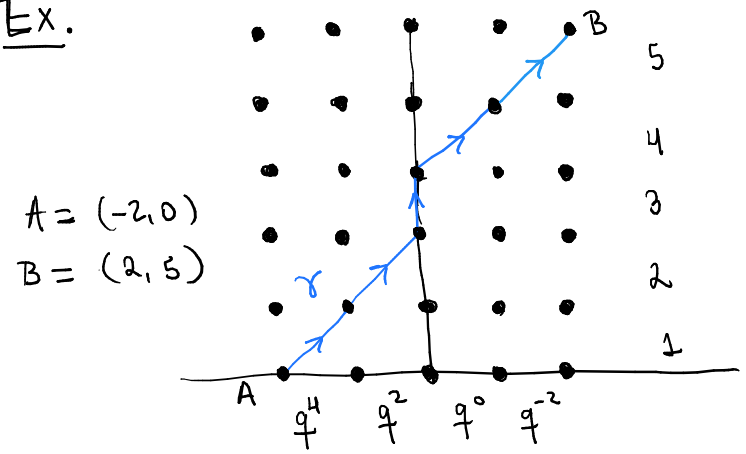
5. A rising path in $\mathbb{Z}\times\mathbb{Z}$ is a finite sequence of steps $\gamma = (s_1, s_2, s_3, \dots)$, where each step s_i is of unit length in the northward direction (N) or of length $\sqrt{2}$ in the northeastward direction (NE). If γ starts at A and end at B, we write $A \xrightarrow{\gamma} B$ or $\gamma : A \to B$.



For a rising path γ, let D_γ be the set of the NE diagonal steps occuring on γ. For $d \in D_\gamma$, let d_x be the x-coordinate of the initial point of d and d_y be the y-coordinate of the terminal point of d. We associate to γ the monomial $\Lambda_a^\gamma \in \mathbb{Z}\Lambda$:

$$\Lambda_a^\gamma := \prod_{d \in D_\gamma} \Lambda_{d_y,\, a q^{-2d_x}}.$$

Ex.



$$\Lambda_a^\gamma = \Lambda_{1,\, a q^4} \Lambda_{2,\, q^2} \Lambda_{4,\, a} \Lambda_{5,\, a q^{-2}}.$$

∎

lemma. We have $e_{k,\, a q^{-2r}} = \sum_{(r,0) \xrightarrow{\gamma} (r+k,\, n+1)} \Lambda_a^\gamma$.

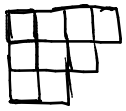
Pf. Any rising path $(r,0) \xrightarrow{\gamma} (r+k, n+1)$ must have k diagonals occuring at the heights $i_1 < i_2 < \dots < i_k$ and starting at the x-coordinates $r < r+1 < \dots < r+k-1$. Hence, $\Lambda_a^\gamma = \Lambda_{i_1,\, a q^{-2r}} \Lambda_{i_2,\, a q^{-2r-2}} \dots \Lambda_{i_k,\, a q^{-2r-2(k-1)}}$.

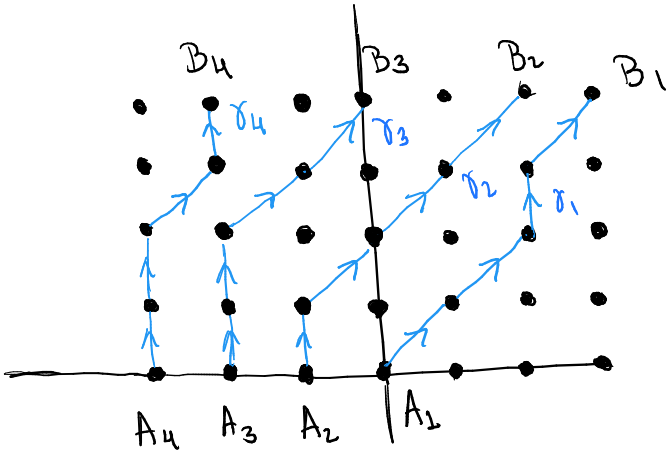
By summing over all such paths, the result follows. ∎

Let λ be a partition and $\ell' = \ell(\lambda')$. For $i = 1, \dots, \ell'$, let

$$A_i = (1-i, 0), \quad B_i = (\lambda'_i + 1 - i, n+1).$$

Given an ℓ'-tuple of non-intersecting paths $\theta = (\gamma_1, \dots, \gamma_{\ell'})$ with $\gamma_i : A_i \to B_i$, let T_θ be the tableau of shape λ in which we fill the i-th column of λ with the weights of the diagonals occuring in γ_i from top to bottom in increasing order.

Ex. $n+1 = 4$, $\lambda =$ 



$w = (\gamma_1, \gamma_2, \gamma_3, \gamma_4) \mapsto T_\theta =$

1	2	3	3
2	3	4	
4	4		

Lemma. The assignment $\theta \mapsto T_\theta$ gives a bijection between $SSYT(\lambda, S)$ and the set of ℓ'-tuples of non-intersecting paths with $\gamma_i : A_i \to B_i$. Moreover,

$$\prod_{i=1}^{\ell'} \Lambda_a^{\gamma_i} = \Lambda_a^{T_\theta}.$$

Pf. T_θ strictly increases in column by construction. Moreover, the non-intersection condition implies that the j-th diagonal of γ_i cannot be higher than the j-th diagonal of γ_{i+1}, so that T_θ weakly increases in rows. Thus, $T_\theta \in SSYT(\lambda, S)$. The inverse map $T \mapsto \theta_T$ is obtained by reading off from the i-th column of T the weights of the diagonals occuring on the i-th path $A_i \to B_i$, in order.

Now, for $\prod_{i=1}^{\ell'} \Lambda_a^{\gamma_i} = \Lambda_a^{T_\theta}$, we just need to notice that the x-coordinate of the first diagonal of γ_i is $x = -(i-1)$, which corresponds to the shift by $q^{2(i-1)}$ given to the i-th column of T_θ. Then, as we move along the diagonals of γ_i by shifts of q^{-2}, that also corresponds to the shifts by q^{-2} given to the entries of the i-th column of T_θ from top to bottom. ∎

Now, we state our main result:

Thm Let $\ell = \ell(\lambda)$, $\ell' = \ell(\lambda')$. We have:

$$\det\left(e_{\lambda_i' - i+j}, a q^{-2(i-j)}\right)_{\ell' \times \ell'} = \sum_{T \in SSYT(\lambda, S)} \Lambda_a^T = \det\left(h_{\lambda_i - i+j}, a q^{2(\lambda_i - i)}\right)_{\ell \times \ell}$$

In particular, if $\ell \leq n$, the above expresses $\chi_q(V(\lambda, a))$ as a polynomial in the q-characters of fundamental modules, as well as a polynomial in the q-characters of symmetric powers of the 1st fundamental module.

The result will follow as an application of the LGV lemma.

LGV lemma Let $A = \{A_1, \dots, A_k\}$ and $B = \{B_1, \dots, B_k\}$ be a set of vertices of a directed graph G. Suppose that the set of all paths $A_i \to B_j$ forms a finite acyclic graph and let w be a weight function on such paths. Let $M_{ij} = \sum_{\gamma : A_j \to B_i} w(\gamma)$. Then,

$$\det(M_{ij}) = \sum_{\theta = (\gamma_1, \dots, \gamma_k)} \operatorname{sgn}(\theta) \prod_{i=1}^{n} w(\gamma_i),$$

where the sum is over the set all k-tuples of non-intersecting paths θ and $\operatorname{sgn}(\theta)$ is the sign of $\sigma \in S_k$ so that $\gamma_i : A_i \to B_{\sigma(i)}$ $\forall$ i.

Pf of the Theorem.

Take G to be the directed acyclic graph with vertex set $\mathbb{Z}\times\mathbb{Z}$ and edges being the (N) step or (NE) step between vertices.

Set $A_i = (1-i, 0)$, $B_i = (\lambda_i' + 1 - i, n+1)$ for $i = 1, \dots, \ell' = \ell(\lambda')$. For a path $\gamma: A_i \to B_j$, let $w(\gamma) := \Lambda_a^\gamma$ as before. From a previous lemma, we have

$$M_{ij} = \sum_{\gamma:(1-j,0)\to(\lambda_i'+1-i,n+1)} \Lambda_a^\gamma = e_{\lambda_i'-i+j,\, a q^{-2(1-j)}}.$$

Then, from the LGV lemma we get

$$\det\left(e_{\lambda_i'-i+j,\, a q^{-2(1-j)}}\right) = \sum_{\theta=(\gamma_1,\dots,\gamma_{\ell'})} \operatorname{sgn}(\theta) \prod_{i=1}^{\ell'} \Lambda_a^{\gamma_i},$$

where the sum is over ℓ'-tuples of non-intersecting paths. However, notice that from our choice of $A = \{A_1, \dots, A_{\ell'}\}$ and $B = \{B_1, \dots, B_{\ell'}\}$, any θ satisfying this condition must be $\theta = (\gamma_1, \dots, \gamma_{\ell'})$ so that $\gamma_i: A_i \to B_i$. Since all such ℓ'-tuples are in bijection with $SSYT(\lambda, S)$, the RHS equals $\sum_{T\in SSYT(\lambda,S)} \Lambda_a^T$.

∎

Rmk. The identity $\sum_{T\in SSYT(\lambda,S)} \Lambda_a^T = \det\left(h_{\lambda_i - i + j,\, a q^{2(\lambda_i - i)}}\right)_{\ell\times\ell}$ has an analogous proof, in which we take paths with (N) step or (E) step. To a horizontal edge h, we associate the monomial $\Lambda_{hy,\, a q^{-2hx}}$, where hy is the y-coordinate $+1$ of the horizontal line containing h and hx is the x-coordinate of its endpoint.

6. Recall that the isomorphism classes of finite-dimensional irreducible representations of $\mathfrak{gl}_{n+1}$ are in bijection with the set of partitions $\lambda = (\lambda_1, \dots, \lambda_{n+1})$, $\lambda_i \geq 0$.

$$\lambda \longleftrightarrow [V(\lambda)].$$

Consider the embedding $\mathfrak{gl}_n \hookrightarrow \mathfrak{gl}_{n+1}$ given by top-left inclusion of matrices. We have the following lemma:

Lemma Let $\lambda = (\lambda_1, \dots, \lambda_{n+1})$. As a $\mathfrak{gl}_n$-module, we have the decomposition:

$$V(\lambda) = \bigoplus_{\lambda/\mu \text{ horizontal strip}} V(\mu), \qquad \mu = (\mu_1, \dots, \mu_n).$$

↳ each column of λ/μ has at most 1 box.

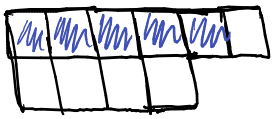
Pf. Since characters commute with restrictions, it follows that:

$$\chi\left(V(\lambda)\big|_{\mathfrak{gl}_n}\right) = s_\lambda(x_1, \dots, x_n, 1).$$

Since $s_\lambda = \sum_T x^T$ with $T \in SSYT(\lambda, S)$, then $s_\lambda(x_1, \dots, x_n, 1) = \sum_{T'} x^{T'}$, where T' is the semi-standard tableau obtained from T by removing all boxes filled with $n+1$. Now, T' has shape μ such that λ/μ is a horizontal strip, so the result follows. ∎

Now, notice that the condition λ/μ is a horizontal strip is equivalent to:

$$\lambda_i \geq \mu_i \geq \lambda_{i+1} \quad \forall i. \qquad \lambda_i \geq \mu_{i+1} \geq \lambda_{i+1}$$



Thus, if we keep further decomposing $V(\mu)$ with respect to $\mathfrak{gl}_{n-1}$ and so on, we will eventually arrive at irreducible representations of $\mathfrak{gl}_1$, which are 1-dimensional. Therefore, we obtain a basis of $V(\lambda)$ parametrized by GZ patterns!

$$\begin{array}{ccccccccc} \lambda_1 & & \lambda_2 & & \lambda_3 & \cdots & & & \lambda_{n+1} \\ & \lambda_1^{(n)} & & \lambda_2^{(n)} & & \cdots & & \lambda_n^{(n)} & \\ & & \ddots & & & & \iddots & & \\ & & & \lambda_1^{(2)} & & \lambda_2^{(2)} & & & \\ & & & & \lambda_1^{(1)} & & & & \end{array}$$

$$\lambda_j^{(i)} \geq \lambda_j^{(i-1)} \geq \lambda_{j+1}^{(i)}$$

$$\begin{aligned} \lambda^{(n+1)} &= \lambda \\ \lambda^{(n)} &= \mu \\ &\vdots \\ \lambda^{(2)} & \\ \lambda^{(1)} & \end{aligned}$$

Ex. $n+1=3$. Then,

$$V(2,1,0) \underset{gl_2}{\simeq} V(2,1)\oplus V(2,0)\oplus V(1,1)\oplus V(1,0)$$

$$\underset{gl_1}{\simeq} \big(V(2)\oplus V(1)\big)\oplus\big(V(0)\oplus V(2)\oplus V(1)\big)\oplus\big(V(1)\big)\oplus\big(V(1)\oplus V(0)\big)$$

The GT pattern $\begin{matrix} 2 & 1 & 0 \\ & 2 & 1 \\ & & 1 \end{matrix}$ for example corresponds to $V(1)\subseteq V(2,1)\subseteq V(2,1,0)$. ◈

For a GT pattern Ω with the j-th row given by the partition $\lambda^{(j)}=(\lambda_1^{(j)},\dots,\lambda_j^{(j)})$, we associate $T_\Omega \in SSYT(\lambda^{(n+1)}, S)$ as follows. We view $\lambda^{(j)}$ as a subdiagram of $\lambda^{(n+1)}$ by identifying top left corners. Then, we fill with the number j the boxes of $\lambda^{(j)}/\lambda^{(j-1)}$. Notice that T_Ω increases in columns by the fact that $\lambda^{(1)}\subseteq\lambda^{(2)}\subseteq\lambda^{(3)}\subseteq\cdots\subseteq\lambda^{(n+1)}$.

Ex. $\Omega = \begin{matrix} 4 & 2 & 2 & 1 \\ & 4 & 2 & 1 \\ & & 3 & 2 \\ & & & 2 \end{matrix} \mapsto T_\Omega =$

1	1	2	3
2	2		
3	4		
4			

Lemma. The assignment $\Omega \mapsto T_\Omega$ gives a bijection between the set of GT patterns of $\lambda^{(n+1)}$ and $SSYT(\lambda^{(n+1)}, S)$.

Pf. The assignment is clearly injective and both sets have the same cardinality $\#GT = \dim V(\lambda^{(n+1)}) = \#\,SSYT(\lambda^{(n+1)}, S)$ ▣